



Cool Flame Investigation (CFI)

SCIENCE FINAL REPORT



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7-23-2020

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0. Nomenclature

CFI	Cool Flame Investigation (ISS droplet combustion experiment focused on cool flames on normal alkane fuels)
CIR	Combustion Integrated Rack
FLEX	Flame Extinguishment Experiment (ISS droplet combustion experiments focused on spacecraft fire safety)
FLEX-2	Flame Extinguishment Experiment - 2 (ISS droplet combustion experiments focused on fundamental aspects)
FOV	field of view
ISS	International Space Station
MDCA	Multi-used Droplet Combustion Apparatus
NTC	Negative Temperature Coefficient
OH*	hydroxyl radical
TSC	Tele-science Support Center

1. Overview

The CFI experiments are conceived as follow-up experiments to the FLEX and FLEX-2 droplet combustion experiments onboard the ISS to further explore the newly discovered quasi-steady “cool flames.” Cool flames are controlled by low temperature chemistry, and occur in the NTC region of n-alkane fuels. FLEX series of experiments showed for the first time that quasi-steady stable flames can be established surrounding a droplet in microgravity under certain conditions, without a visible flame at temperatures in the range of 700 to 900 K, unlike the hot flames where the flame temperature is higher than 1200 K. Since the cool-flame chemistry depends on the chemical structure of the fuel as well as pressure and temperature, three different fuel combinations were investigated, namely, pure n-dodecane, n-dodecane/iso-dodecane mixtures and farnesane (a renewable jet fuel) at ambient pressures up to 5 atmospheres. Cool flame extinction occurs when its static stability is lost close to the lower turnover point in the NTC region, addition of helium to the ambient gas is also considered. Results for droplet burning rates, flame dynamics, droplet extinction diameters, and flame radiant out put are measured.

2. Introduction (PS):

Understanding the combustion characteristics of liquid hydrocarbon fuels is of practical and fundamental importance. The United States alone consumes approximately 19 million barrels of liquid fuels per day which averages out to an astonishing 2.5 gallons per day per person (Reitz, 2013). The primary reasons for such a high reliance on hydrocarbon fuels as the single most important energy source, not only in the US but globally, is their high energy density, ease of handling, and portability compared to many other forms of energy. With the emergence of newer internal combustion engine technologies such as Homogeneous Charge Compression Ignition (HCCI) and Reactivity-Controlled Compression Ignition (RCCI) engines, understanding the autoignition and heat release chemistry and kinetics of hydrocarbon fuels is more important than ever. These new technologies coupled with the emergence of major sources of renewable fuels for both gasoline and diesel applications, including butanols and heavier alcohols, bio-diesel, farnesane and high carbon number synthetics, further emphasize the need to understand the kinetic effects of blending renewables with petroleum derived fuels.

a. Objectives

Cool flame phenomena are influenced by a number of factors, including the fuel molecular structure (for pure fuel cases), the ambient environment (pressure, temperature, oxygen and diluent (He) concentrations), and the fuel composition (the mixture of fuel structures and perhaps fuel additives such as those used to change octane or cetane behavior). Though it is not possible to examine all of these factors in a thorough manner, here we propose a set of experiments that explore each one of these factors, namely, fuel structure effects, ambient conditions effect, fuel mixture effects, and the presence of fuel additives.

b. Team Members

Science Definition Team:

Prof. Forman A. Williams, University of California, San Diego

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Daniel L. Dietrich, Project Scientist

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c. Dates of Operation

CFI experiments were conducted onboard the ISS from January 2017 through September 2017.

3. Science Overview

a. Background

Understanding the combustion characteristics of liquid hydrocarbon fuels is of practical and fundamental importance. The United States alone consumes approximately 19 million barrels of liquid fuels per day, which averages out to an amazing 2.5 gallons per day per person. The primary reason for such high reliance on hydrocarbon fuels as the single most important energy source, not only in the US but in the rest of the world, is their high energy density, ease of handling, and portability, compared to many other forms of energy sources. With the emergence of newer technologies for internal combustion engines and the increasing use of renewable bio-fuels, such as the Homogeneous Charge Compression Ignition (HCCI) engine and bio-diesel, that promise cleaner more efficient and sustainable utilization of hydrocarbon fuels, the need to understand the fundamental chemical and physical aspects of liquid-fuel combustion is bound to increase.

Commercially available transportation fuels, such as gasoline, diesel, and jet fuel, are comprised of many hydrocarbon compounds, normal alkanes and branched alkanes being major components. The combustion characteristics of these alkanes are quite complex and follow branched-chain reaction pathways involving many intermediate radicals. The chemical pathways vary, depending on temperature and, to a lesser extent, on pressure. In general it is possible to identify a low-temperature reaction mechanism and a high-temperature mechanism, with a transition region between them. Many hydrocarbons exhibit a unique feature in this transition region, namely that the reaction rate decreases with increasing temperature,

leading it to be called the negative-temperature-coefficient (NTC) region. There are a number of models available in the literature, simplified chemical-kinetic model involving hundreds of reactions as well as detailed mechanisms involving thousands of reactions.

Droplet-combustion experiments (FLEX, FLEX-2) conducted onboard the International Space Station, as part of the Flame Extinguishment Experiments, have revealed for the first time that large n-alkane droplets can continue to burn quasi-steadily in the NTC region (see, Fig 1). NTC behavior of the heat-release rate is necessary for low-temperature quasi-steady droplet burning to be stable. It was observed that the second-stage combustion of the droplet controlled by low-temperature NTC chemistry, referred to as "cool-flame combustion" here for the sake of simplicity, abruptly ends at a finite droplet diameter, leaving the unburned droplet to vaporize in the hot atmosphere. A vapor cloud was also found to form around the droplet as the partially oxidized fuel condensed immediately following cool-flame extinction.

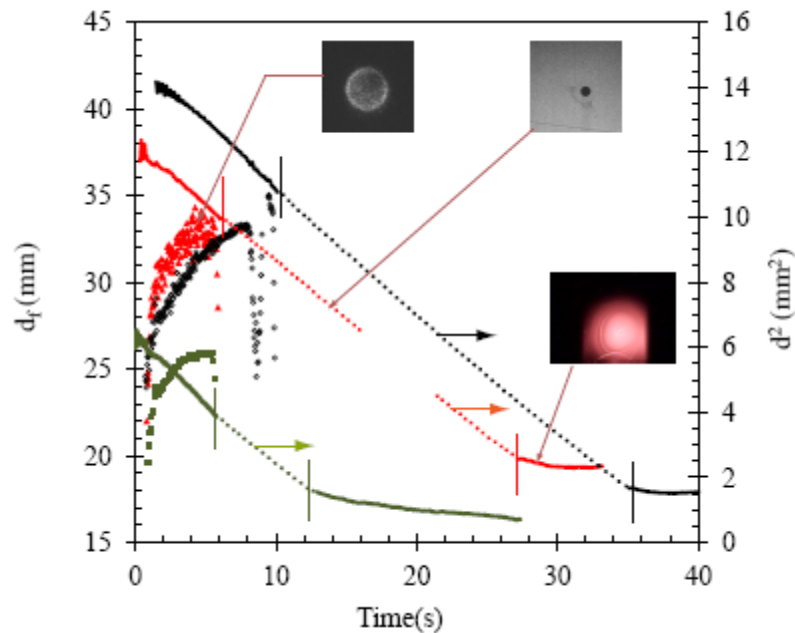


Figure 1. Three representative examples of radiative extinction and subsequent cool flame combustion of n-heptane droplets in air. Inset shows the visible flame, backlit droplet images and the vapor cloud formation.

Furthermore, the second-stage flame is shown to lie closer to the droplet surface where the rates of heat generation by the NTC chemistry is dynamically balanced by the heat loss rate. Unlike the hot-flame combustion where the overall process is controlled by equilibrium thermodynamic properties of the fuel, the cool-flame combustion depends on the chemical-kinetic details of the NTC region. These experimental and numerical studies provide a new tool to further explore the NTC chemistry of liquid fuels in an entirely new way. The experimental cool-flame droplet burning rates provide measures of the heat-release rates and stoichiometry, the flame-diameter evolution provides insight into hot-flame diffusive and convective heat-loss effects, and the droplet extinction diameter provides a stringent measurement of chemical-kinetic characteristic times.

The objectives of the CFI experiments is to further explore the cool flame phenomenon using pure fuels, and fuel mixtures over a range of pressures and helium dilution.

b. Rational for Microgravity (if unusual)

Rationale for microgravity remains the same for all the droplet combustion experiments conducted onboard the ISS. In microgravity the droplets burn in a spherically symmetric form, and therefore it is possible to model the processes with complex chemical kinetic mechanisms involving thousands of reactions leading to a better understanding of the chemistry. Moreover, the cool flames need much longer residence times and cannot be observed under normal gravity conditions with droplet flames.

c. NRA the Investigation was Selected Under

This project was initiated by NASA through a Science Definition Team (SDT) which formulated the overall objectives. It was peer reviewed by an independent panel of scientists and approved by NASA HQ as a follow-up experiments to FLEX, and FLEX-2 in the CIR using additional fuels and gases needed to complete the test matrix. The experimental hardware is same as that of the FLEX experiments with few minor modifications in the data acquisition systems.

d. Final Experiment Objectives

Examine the cool flame combustion behavior of n-dodecane, n-dodecane/iso-dodecane fuel mixtures, and farnesane fuels over a range of pressure from 0.25 atm to 5 atm.

4. Operations

a. Procedures

The operational procedure used in the CFI tests is the same as for the FLEX experiments. Once the ISS crew installs the required fuel syringe, and the gas bottles containing the gas mixtures in the MDCA and CIR facilities, all the actual experimental operations are conducted from the ground at the Telescience Support Center (TSC) located at the NASA Glenn Research Center. Prior to each test the combustion chamber is filled with the specified gas mixture at the required pressure, and the hot-wire igniters are positioned close to the droplet deployment site. Subsequently, a droplet of desired size is dispensed between the hypodermic needles and freely deployed, and ignited. The igniter is withdrawn from the FOV once ignition is detected. When required a 40 micron silicon-carbide fiber is used to tether the droplet at one place and stop it from drifting off from the field of view.

The combustion process is recorded using a backlit, black-and-white camera (5 cm X 5 cm FOV) to obtain droplet size as a function of time, a UV-sensitive camera is used to image the OH* emission from the flame which provides the flame size data. A color camera which is had a live downlink to the TSC is used to perform and monitor the overall process. Narrow-band and wide-band radiometers are used to measure the radiation emission from the flame. Details of the experimental hardware and the data reduction procedures can be found in Dietrich et al. (2014)

b. On-Orbit Testing

Over two hundred tests were carried out during the CFI experimental campaign. The initial droplet size was kept around 4 mm so that radiative extinction and the subsequent cool flame combustion can be observed. The list of experiments performed are as follows:

- Pure n-dodecane at 0.5, 0.75, 1, 1.25, 2, 3, and 5 atm.
- N-dodecane/Iso-dodecane mixtures: 75/25% and 60/40% by mass at pressures from 0.5 to 5 atm
- Helium dilution by 29% and 59% with O₂ mole fraction at 0.21
- Farnesane: from 0.75 to 3 atm

c. Anomalies That Effect Science Surprises/New Discovery

No new anomalies or surprises were observed during the CFI experiments. However, as the ambient pressure is increased above 2 atm, multiple reignition cycles were observed, where first the hot flame extinguishes due to radiative heat loss followed by cool flame, and then reignition to a hot-flame followed by a cool flame, with this cycle repeating several times until the fuel droplet is completely consumed. This behavior is unexpected based on earlier observations from FLEX experiments. Furthermore, the soot-shell

which surrounds the droplet and lies between the droplet surface and the flame, became unstable and ill-defined at higher pressures, a phenomenon which is not fully understood.

The organization of the data posted in the PSI website is as follows:

5. Results

All the CFI data is yet to be fully analyzed and interpreted using theoretical and numerical models. A brief overview of general observations made during **preliminary** data analysis prior to posting on the PSI website is presented below. Interested researchers are welcome to propose to NASA for further analysis and interpretation of CFI data.

Cool flames were observed during n-dodecanol combustion as well as n-dodecanol/iso-dodecanol mixture cases. The addition of iso-dodecanol reduces the tendency to form cool flames in n-dodecanol. Figure 1 shows a typical case of cool-flame behavior for n-dodecanol.

Figure 1. FLEX-1223: N-dodecane at 1.25 atm pressure in air.

Warm flames are a consequence of changing in chemical pathway during n-alkane oxidation. Figure 2 shows (Westbrook, 2010) the various chemical pathways: cool flames occur when low-temperature branching is dominant, when intermediate temperatures become more active warm flames are established. During warm flames the heat release rate dynamically balance the heat loss rate. Addition of helium seems to enhance this balance within the NTC region.

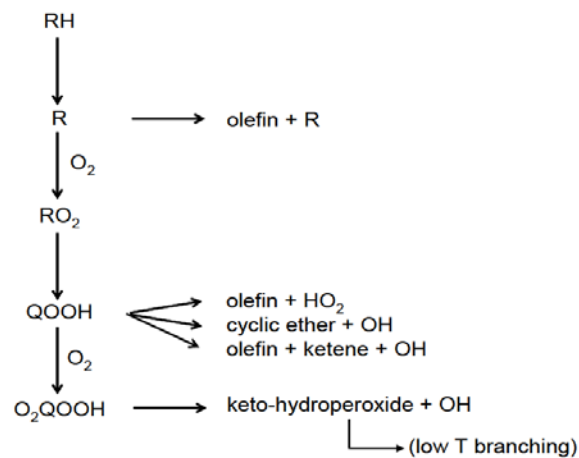


Figure 2. Chemical kinetic pathways for n-alkane oxidation (Westbrook, 2010).

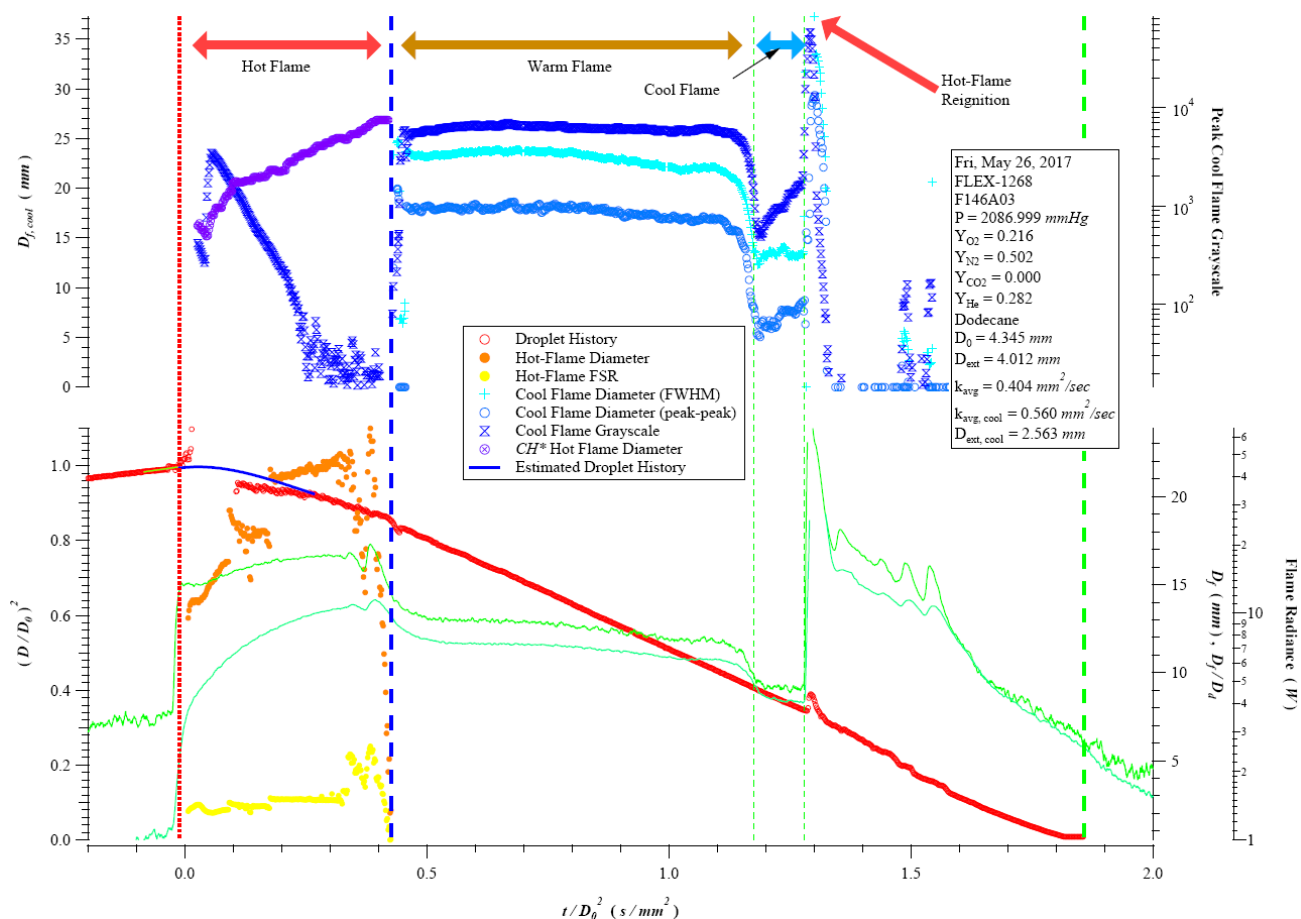


Figure 3: FLEX-1268 – N-dodecane at 3 atm burning in 12%O₂ – 50% N₂ – 29% Helium. Reignition and warm flame formation are shown.

Warm flames were observed typically at higher pressures following radiative extinction and the flame is not visible in the color camera. Subsequently, a hot flame is ignited which quickly extinguishes to form a cool flame.

5.3 Cool flame extinction:

Cool flame extinction is controlled primarily by chemical kinetics. As the droplet size decreases, the residence time in the reaction zone of the gas-phase diffusion flame is reduced to a value corresponding to which the cool flame becomes statically unstable, and this leads to cool flame extinction. Figure 4 shows the cool flame extinction diameter as function of ambient pressure for both free-floated and fiber-supported droplets for pure n-dodecane burning in air. As mentioned earlier, as the pressure is increased above approximately 2 atm, reignitions were commonly observed, eventually leading to complete combustion of the fuel droplet. It is interesting to note that the 40 micron fiber shows relatively little effect on the extinction diameter.

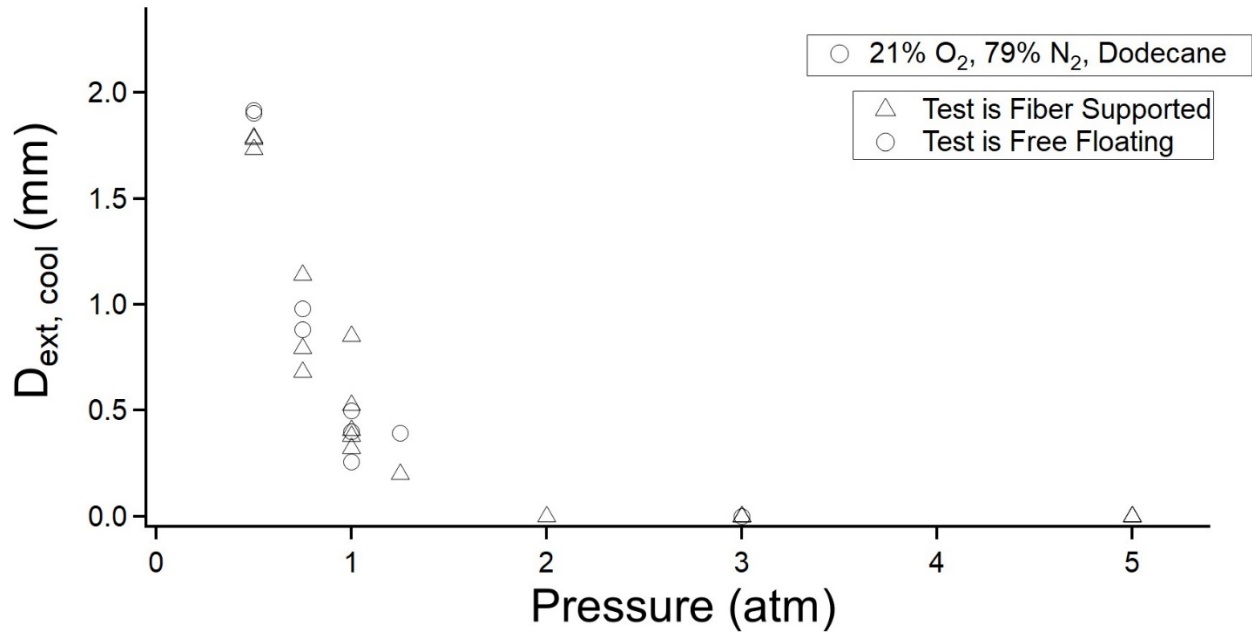


Figure 4. Cool flame extinction diameters as a function of ambient pressure for n-dodecane droplets burning in air.

Since the cool flame extinction is an instability phenomenon where the dynamic balance between heat release and heat loss from the flame is lost at the lower turning point in the NTC region, one would expect addition of helium to have a substantial influence on the process. Figure 5 illustrates the effect of helium dilution on the extinction diameter for n-dodecane. Increase helium dilution leads to earlier cool flame extinction giving a larger extinction diameter.

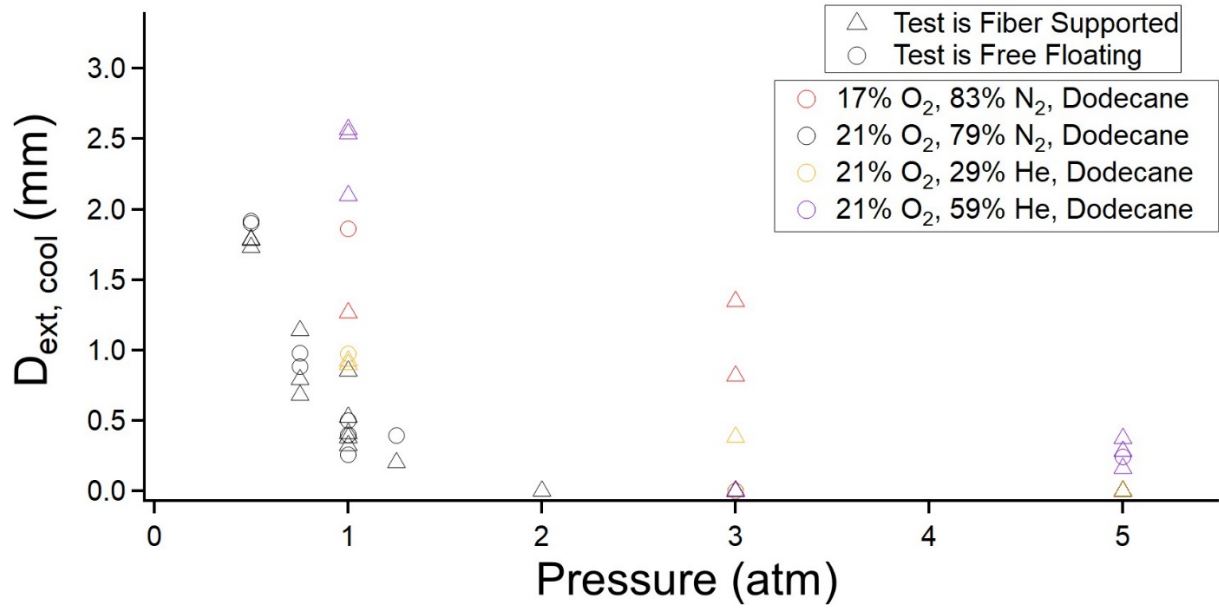


Figure 5. Effect of helium dilution on cool flame extinction during n-dodecane droplet combustion.

5.4 Sooting behavior as function of ambient pressure:

The effects and the process of soot formation during cool flame combustion remains largely unexplored. The soot shell that forms during hot flame is stabilized by the balance of force between fluid mechanical drag and the thermophoretic force. At low pressures a stable spherical soot-shell is formed. However, as the pressure increases the soot-shell becomes diffuse and undefined. The reasons for this phenomenon is still not well understood. At low pressures, due to the low flame temperature of the cool flame, soot is not formed. The soot that is formed during the hot flame combustion is simply convected away from the droplet surface. But, when the pressure increases, residual soot is seen surrounding the droplet even after the hot flame extinction. The mechanisms behind this process has not been fully understood, and further analysis and research is needed.

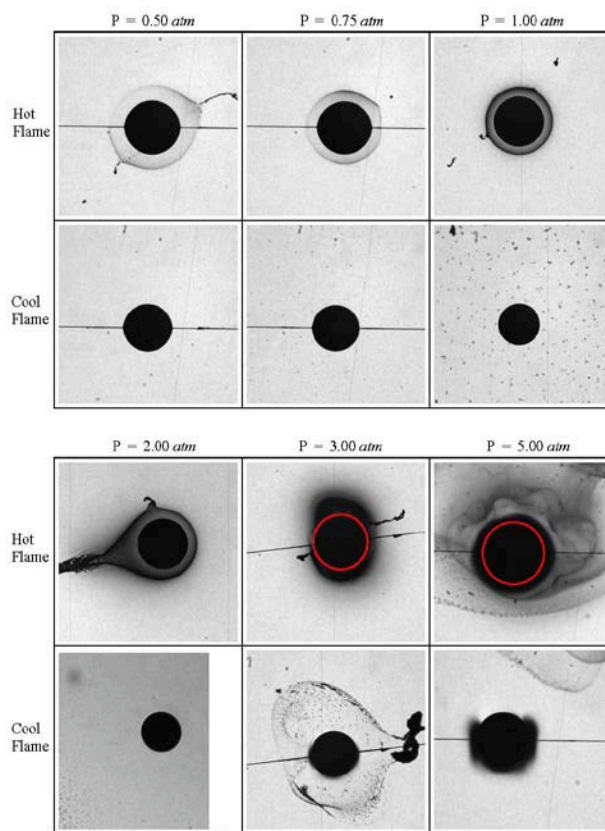


Figure 6. Backlit images of the droplet during hot and cool flame burning at atmospheric pressures of approximately 380, 570, 760, 1520, 2280 and 3800 mmHg. The hot flame images are approximately 4 sec after igniter withdraw and the cool flame images are approximately 5 sec after hot flame extinction. The droplet in the cool flame image at 1520 mmHg was close to the edge of the FOV. The red circles during hot flame burning at the two highest pressures show the approximate location of the droplet surface.

5.5 Summary of findings:

CFI experiments has provided cool flame burning data for pure n-dodecane, and n-dodecane/iso-dodecane mixtures. Farnesane, a commercially available renewable fuel characteristics were also studied. The effect of pressure and helium dilution were studied for these fuels. Data for droplet regression rate, hot and cool flame diameters, extinction conditions, and radiant emission, both broad band and narrow band, are obtained and posted to the PSI website.

b. Benefits

Low temperature chemistry is of fundamental importance in the combustion of many practical fuels. Low temperature chemistry controls the autoignition of fuels when injected into a combustion chamber, and in homogeneous mixtures of fuel and oxygen. New internal combustion engines, such as the HCCI engines, operate at low temperatures reducing the pollution formation and increasing overall efficiencies. The CFI experimental measurements provide a useful way to validate the complex chemical kinetic mechanisms that are being developed. The cool flame combustion process itself maybe of use in fuel reformulation technologies to produce hydrogen in fuel cell applications.

c. Patents

None

6. Appendices

a. 30-Day Operational Accomplishments Report

Not available

b. Publications

The following publications are from the FLEX series of experiments and address various aspects of cool flame supported droplet combustion. They are listed here because they form the motivation for the CFI experiments. A limited data from CFI experiments pertaining to cool flame extinction diameters are published and included in this list.

Dietrich, Daniel L., Vedha Nayagam, Michael C. Hicks, Paul V. Ferkul, Frederick L. Dryer, Tanvir Farouk, Benjamin D. Shaw et al. "Droplet combustion experiments aboard the international space station." *Microgravity Science and Technology* 26, no. 2 (2014): 65-76.

Farouk, T., & Dryer, F. L. (2011). Microgravity droplet combustion: effect of tethering fiber on burning rate and flame structure. *Combustion Theory and Modelling*, 15(4), 487-515.

Farouk, Tanvir I., and Frederick L. Dryer. "Isolated n-heptane droplet combustion in microgravity: "Cool Flames"—two-stage combustion." *Combustion and Flame* 161.2 (2014): 565-581.

Farouk, T. I., Hicks, M. C., & Dryer, F. L. (2015). Multistage oscillatory "Cool Flame" behavior for isolated alkane droplet combustion in elevated pressure microgravity condition. *Proceedings of the Combustion Institute*, 35(2), 1701-1708.

Farouk, T. I., Dietrich, D., Alam, F. E., & Dryer, F. L. (2017). Isolated n-decane droplet combustion—Dual stage and single stage transition to "Cool Flame" droplet burning. *Proceedings of the Combustion Institute*, 36(2), 2523-2530.

Farouk, T. I., Dietrich, D., & Dryer, F. L. (2019). Three stage cool flame droplet burning behavior of n-alkane droplets at elevated pressure conditions: Hot, warm and cool flame. *Proceedings of the Combustion Institute*, 37(3), 3353-3361.

Liu, Y. C., Xu, Y., Avedisian, C. T., & Hicks, M. C. (2015). The effect of support fibers on micro-convection in droplet combustion experiments. *Proceedings of the Combustion Institute*, 35(2), 1709-1716.

Nayagam, V., Dietrich, D. L., Ferkul, P. V., Hicks, M. C., & Williams, F. A. (2012). Can cool flames support quasi-steady alkane droplet burning?. *Combustion and Flame*, 159(12), 3583-3588.

Nayagam, V., Dietrich, D. L., Hicks, M. C., & Williams, F. A. (2015). Cool-flame extinction during n-alkane droplet combustion in microgravity. *Combustion and Flame*, 162(5), 2140-2147.

Nayagam, V., Dietrich, D.L. and Williams, F.A., 2017. A Burke-Schumann analysis of dual-flame structure supported by a burning droplet. *International Communications in Heat and Mass Transfer*, 87, pp.84-89.

Seshadri, K., Peters, N., Williams, F. A., Nayagam, V., & Paczko, G. (2016). Asymptotic analysis of quasi-steady n-heptane droplet combustion supported by cool-flame chemistry. *Combustion Theory and Modelling*, 20(6), 1118-1130.

Shaw, B. D. (2014). ISS droplet combustion experiments-uncertainties in droplet sizes and burning rates. *Microgravity Science and Technology*, 26(2), 89-99.

Shaw, B. D., & Vang, C. L. (2016). Oxygen Lewis number effects on reduced gravity combustion of methanol and n-heptane droplets. *Combustion Science and Technology*, 188(1), 1-20.

Tyurenkova, V. (2019). Two regimes of a single n-heptane droplet combustion. *Acta Astronautica*, 163, 25-32.

Paczko, G., Peters, N., Seshadri, K., & Williams, F. A. (2014). The role of cool-flame chemistry in quasi-steady combustion and extinction of n-heptane droplets. *Combustion Theory and Modelling*, 18(4-5), 515-531.

Vang, C. L., & Shaw, B. D. (2018). ISS Experiments on Combustion of Heptane/Hexadecane Droplets. *AIAA Journal*, 56(12), 4858-4869.

Williams, F. A., & Nayagam, V. (2019). Quasi-steady combustion of normal-alkane droplets supported by Cool-Flame chemistry near diffusive extinction. *Combustion Theory and Modelling*, 23(4), 748-770.

Williams, F.A. and Nayagam, V., 2020. Cool-flame dodecane-droplet extinction diameters. *Combustion and Flame*, 212, pp.242-244.

Xu, Y., Farouk, T. I., Hicks, M. C., & Avedisian, C. T. (2020). Initial diameter effects on combustion of unsupported equi-volume n-heptane/iso-octane mixture droplets and the transition to cool flame behavior: Experimental observations and detailed numerical modeling. *Combustion and Flame*, 220, 82-91.

c. Presentations

Dietrich, D. L., Nayagam, V., and Williams, F. A., 2018. Hot and Cool-Flame Combustion of N-dodecane Droplets at Moderately High Pressures in Microgravity, 34th Annual Technical Meeting, The American Society for Gravitational and Space Research, 31 October – 03 November, 2018, Rockville, Maryland.

d. Webpages

<https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/mdca/experiments-using-the-mdca/cfi/>

References to commercial products are made for clarity and do not indicate an endorsement on the part of NASA or the federal government.

This document includes many internet links, where efforts were made to verify that the webpages and other online content was functional when the report was prepared. However, there is of course no guarantee of their continued function.